

ANKIT KUMAR

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Northeastern University, Boston, MA

PROFESSIONAL SUMMARY

Applied Scientist and PhD candidate with demonstrated sustained excellence spanning 12+ years in formal methods, security analysis, and software verification.

Track record includes national-level industry recognition (Droidcon India Hackathon 2013, solo winner) through current academic achievements (Best Student Paper Awards 2023, 2025). Discovered and reported CVE-2022-47547 affecting Ethereum 2.0 networks, resulting in protocol-level patches securing \$100B+ in assets. Author of 7 peer-reviewed publications (32 citations, h-index: 4) with consistent recognition at premier venues including IEEE S&P 2024.

Holds 3 pending USPTO patents from Amazon internships. Recognized internationally through invited presentations at IPFS Camp Lisbon and multiple technical conferences, with peer review service for premier venues (FMCAD, ICFEM, NFM, OOPSLA, ICFP). Pattern of sustained excellence validated by market: \$415,863 compensation placing in top 5-10% of Applied Scientists nationally.

COMPENSATION & MARKET RECOGNITION

- **Amazon Applied Scientist Position (2026):** Total compensation of \$415,863 USD annually (year one)
- Market analysis demonstrates this compensation places candidate in **top 5-10% of Applied Scientists nationally**, significantly exceeding industry standards for recent PhD graduates
- Compensation is 65-110% above typical Applied Scientist base compensation, reflecting specialized expertise in formal methods, security analysis, and agentic AI systems
- Represents recognition of extraordinary ability in software verification and protocol security at enterprise scale

PROFESSIONAL EXPERIENCE

Applied Scientist Intern • Amazon Web Services • Santa Clara, CA • May 19, 2025 - Aug 22, 2025

- Pioneered **Agentic AI methodology** to automatically generate correct Rust code specifications from Java implementations
- Leveraged advanced **dataflow analysis** to eliminate hallucinations and improve code correctness
- Secured **two patent submissions (pending)** for innovative code verification and analysis techniques; earned return offer recognition in a full-time role
- Technologies: Java, Rust, Claude, Cline, Amazon Q, Agentic AI

Member of Technical Staff, Intern • Rivos Inc. • Portland, OR • Jun 7, 2023 - Sep 1, 2023

- Developed formal verification framework for **RISC-V hypervisor page walk algorithm security properties**
- Accelerated proof verification by 10x using strategic lemma transfer from high-level models
- Technologies: ACL2, ACL2 Sedan, Jaspergold

Applied Scientist Intern • Amazon • Minneapolis, MN • May 31, 2022 - Sep 2, 2022

- Engineered **taint-analysis framework** for enterprise-scale tracking of sensitive data across AWS infrastructure

- Secured **one patent submission (pending)**: "Data Flow Analysis of Cloud-Based Software Applications"; earned return offer recognition
- Technologies: Java, Grails, GitHub, Shell Scripting

Software Development Engineer II • Olacabs • Bengaluru, India • 2012-2014

- Architected inaugural Olacabs Android consumer application, capturing **80% of total cab bookings**
- Reduced driver app crashes by **80%** through systematic debugging and optimization
- Implemented robust cashless payment and digital wallet systems
- Technologies: Java, Android, Git, Ruby on Rails

Associate • Morgan Stanley Advantage Services • Mumbai, India • 2010-2012

- Transformed legacy derivatives trading platform for multi-regional deployment
- Expanded platform reach to Hong Kong and Singapore, **doubling transaction capacity**
- Technologies: .NET, Java, Spring, Perforce

Upcoming Role: Applied Scientist • Amazon Web Services • Santa Clara, CA • Starting Jan 2026

EDUCATION

PhD in Computer Science • Northeastern University • Boston, MA • 2018-Expected 2026

- Thesis: Refinement-based reasoning of P2P protocols is feasible and useful
- **Research Impact:** 7 peer-reviewed publications, 32 citations, h-index: 4, i10-index: 1

PhD Research Contributions

- **1. P2P Protocol Verification**
 - Applied formal methods and refinement theory to **Gossipsub/Floodsub protocols**
 - Discovered and reported [**MITRE CVE-2022-47547**], a critical scoring function vulnerability affecting Ethereum and Filecoin networks, leading to real-world fixes
 - Published multiple peer-reviewed papers and presented work at IPFS Camp Lisbon 2022 and ACL2 Workshops (2023, 2025)
 - Related publications: [1], [2], [3]
- **2. Education: Automated Student Feedback Systems**
 - Developed **theorem-prover-backed automated feedback systems** for grading student submissions
 - Integrated with Gradescope to provide formal verification of student solutions and proofs
 - Enabled scalable and precise assessment of student work in formal methods courses
 - Related publications: [4], [5], [6]
- **3. Verification tools and algorithms: String Solvers**
 - Designed and implemented sound and theoretically complete **string equation solvers** for verification and formal reasoning tasks
 - Applied these solvers to both academic research and practical software verification problems
 - Related publications: [7]

M.Tech in Computer Science • IIT Kanpur • Kanpur, India • 2015-2017

B.Tech in Electrical Engineering • IIT (BHU) Varanasi • Varanasi, India • 2006-2010

SECURITY RESEARCH & IMPACT

- **CVE-2022-47547 Discovery - Critical Infrastructure Security Impact:**

- Identified and reported critical vulnerability in GossipSub 1.1 protocol, the backbone of Ethereum 2.0's peer-to-peer consensus mechanism
- **Systems Affected:** Ethereum 2.0 Beacon Chain (securing 1M+ validators and \$100B+ in staked assets as of 2024), Filecoin network, and IPFS infrastructure
- **Severity:** CVSS Score 5.3/10 (Medium) - Network-based attack vector with low complexity, affecting message propagation integrity
- **Official Recognition:** Published in National Vulnerability Database (NVD) on December 19, 2022; Last modified April 17, 2025
- **Industry Impact:** Coordinated disclosure with MITRE (CVE Program); resulted in protocol-level patches deployed across Ethereum ecosystem, affecting global blockchain infrastructure
- **Technical Contribution:** Vulnerability allows malicious peers to maintain positive reputation scores while continuously misbehaving, enabling sustained network manipulation attacks
- **Verification Method:** Discovered through formal mathematical modeling and proof-based analysis techniques, demonstrating practical value of formal verification in production systems
- Publication documenting discovery: IEEE Symposium on Security & Privacy 2024 (premier security venue)
- **Formal Verification Research Impact:** Applied mathematical proof techniques to analyze critical peer-to-peer protocols, revealing previously unknown attack vectors in production systems used by millions
- **Academic Contributions:** 7 peer-reviewed publications (32 citations, h-index: 4) demonstrating formal verification methodologies for distributed systems and blockchain protocols
- **Intellectual Property:** 3 patent applications pending with US Patent Office covering novel techniques in dataflow analysis, code verification, and security analysis
 - US Patent Application (Pending): "Data Flow Analysis of Cloud-Based Software Applications"
 - Two additional patent applications (Pending): Novel approaches to AI-driven code generation and verification

PROFESSIONAL SERVICE & COMMUNITY CONTRIBUTIONS _____

- **Peer Review Service:** Reviewed papers for FMCAD 2025, ICFEM 2023, and NFM 2024 (premier formal methods conferences)
- **Artifacts Evaluation:** Served on OOPSLA 2024 and ICFP 2023 Artifacts Evaluation Committees (top-tier programming languages conferences)
- **Conference Leadership:** Invited to chair SIGCSE TS Conference, Canada, 2023 (leading computer science education conference)
- **Volunteer Service:** Student volunteer at FLoC 2022, Haifa, Israel (federated logic conferences)

PUBLICATIONS _____

1. **A Formalization of the Correctness of the Floodsub Protocol**
ACL2 Workshop 2025, Austin, TX • Best Student Paper Award • [\[PDF\]](#)
2. **Formal Model-Driven Analysis of Resilience of GossipSub to Attacks from Misbehaving Peers**
IEEE Symposium on Security and Privacy 2024, San Francisco, CA • [\[PDF\]](#)

3. Verification of GossipSub in ACL2s

ACL2 Workshop 2023, Austin, TX • Best Student Paper Award • [\[PDF\]](#)

4. Automated Grading of Automata with ACL2s

Electronic Proceedings in Theoretical Computer Science 2023 • [\[PDF\]](#)

5. Proving Calculational Proofs Correct

ACL2 Workshop 2023, Austin, TX • [\[PDF\]](#)

6. Calculational Proofs in ACL2s

arXiv preprint 2023 • [\[PDF\]](#)

7. Mathematical Programming Modulo Strings

Formal Methods in Computer Aided Design 2021, Virtual Conference • [\[PDF\]](#)

INVITED TALKS & RESEARCH DISSEMINATION

- **IEEE Symposium on Security and Privacy (S&P) 2024** - Conference Presentation
“Formal Model-Driven Analysis of Resilience of GossipSub to Attacks from Misbehaving Peers”
Premier security conference (top-tier venue), San Francisco, CA
Archived on IEEE Security & Privacy official YouTube channel: [\[Video Link\]](#)
Presented research on CVE-2022-47547 and GossipSub protocol security at leading international security venue
- **IPFS Camp Lisbon 2022** - Invited International Presentation
“Formal Verification of Gossipsub and Network-Level Vulnerabilities”
Hosted by Protocol Labs (creators of IPFS/Filecoin), Lisbon, Portugal
Archived on Protocol Labs official YouTube channel: [\[Video Link\]](#)
Reached international audience through organizer’s platform
- **Syracuse University 2024** - Invited Research Talk
Presented dissertation research on formal methods and P2P protocol verification
- **Boston Computation Club** - Invited Technical Presentation
“Mathematical Programming Modulo Strings”
Archived on Boston Computation Club official channel: [\[Video Link\]](#)
Technical community presentation demonstrating research accessibility and impact

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Rust, Lisp, ACL2, ACL2 Sedan, Coq, Haskell, Ruby
- **Development Tools:** Git, GitHub, Emacs, LaTeX, Shell Scripting, Perforce
- **Frameworks & Technologies:** Spring, .NET, Grails, Ruby on Rails, Android
- **AI & Machine Learning:** Claude, Cline, Amazon Q, Agentic AI
- **Verification Tools:** Jaspergold, ACL2, ACL2 Sedan
- **Specializations:** Formal Methods, Theorem Proving, AI Agents, Software Verification, Security Analysis, Dataflow Analysis, Taint Analysis

AWARDS & HONORS

- **Best Student Paper Award, ACL2 Workshop 2025**
Selected from competitive field of submissions; ACL2 is premier venue for automated reasoning research
- **Dissertation Completion Fellowship, Summer 2024**
Competitive university-wide fellowship recognizing outstanding dissertation research

- **Best Student Paper Award, ACL2 Workshop 2023**

Second Best Student Paper recognition demonstrating sustained research excellence (one of few researchers to receive this honor multiple times)

- **Student Travel Fellowship, PLMW at POPL 2019**

Competitive fellowship for Programming Languages Mentoring Workshop at top-tier conference

- **First Prize, Droidcon India National Hackathon 2013**

National-level Android development competition, Bangalore, India

Project: "Geomessenger" - Solo participant winning first prize among competing teams

Received multiple prizes from event sponsors (Novell, Nokia, Samsung, Perfecto Mobile, Google)

Competition details: [\[Event Page\]](#)

Video presentation: [\[Technical Presentation\]](#) — Award ceremony: [\[Prize Distribution\]](#)

Media coverage: [\[Twitter/X\]](#)